Breeding Diverse Noise: A Population-Based Companion to "It's Never Too Late" Collapse Recovery

Anirud Aggarwal

anirud@vamana.ai research.aniaggarwal.com

Code: github.com/AniAggarwal/divgen (fork of the official implementation)

Abstract

Harrington et al.'s "It's Never Too Late" makes a lovely observation: mode collapse in trained text-to-image models is not a fact about the model so much as a fact about *where we start it* — and the initial noise can simply be optimized, end-to-end, into a diverse set of generations. This note is written in appreciation of that work, and asks one follow-up question of it: does the noise need a *gradient* to find its way? We reformulate their objective as a two-objective *artificial-selection* search[1] over *sets* of initial noises, using the multi-objective machinery from our ECAD work on caching schedules. On GenEval with SDXL-Turbo, breeding reaches the gradient-based optimization's numbers on its own metrics (slightly ahead on three of the four reported, slightly behind on LPIPS) — without a single backward pass through the sampler. We describe what the population view adds (memory-free scaling to large models, non-differentiable set-level objectives, an explicit quality–diversity Pareto front, self-tuned step sizes) and what it costs, and offer it as a companion, not a replacement, to the original method.

1 Appreciation, and a question

Three ideas in Harrington et al. do a lot of work. First, that inference-time diversity is recoverable at all — that a collapsed model still *contains* its diversity, reachable through the initial noise x_0. Second, the frequency analysis (their Fig. 9): optimization acts almost entirely on the lowest third of the noise spectrum, which both explains the method and immediately yields the pink-noise initialization that improves every method they test, including the baselines. Third, the observation that *set-level* objectives (DPP, Vendi) are the right currency for diversity, because they cannot be gamed by making one image strange — a point their user study makes empirically.

The method that operationalizes these ideas is gradient descent through the frozen sampler: differentiable diversity and quality scores are backpropagated to the noise batch. Our question is narrow: *is the gradient essential, or is it one search procedure among several for the landscape their analysis reveals?* The question matters practically: backpropagating through a 10B-parameter flow model is a memory wall on commodity hardware, and the most human-aligned objectives in their study (DPP and Vendi at the set level) are exactly the ones where differentiability is most awkward.

We answer it by transplanting the multi-objective search we built for ECAD [2] — which breeds diffusion *caching schedules* against a quality–speed front — onto their problem: same NSGA-II [3] engine, same two-objective structure, with the chromosome swapped from binary cache decisions to a set of noise tensors.

2 Breeding noise sets

Individuals. Diversity in the source paper is measured over a set of $B{=}4$ images per prompt, so the unit of selection is the whole set: an individual is $\{x_0^{(1)}, \ldots, x_0^{(B)}\}$, plus two *strategy genes* (a mutation step σ and a per-element mutation rate ρ) that are inherited and perturbed alongside the noise, as in classical self-adaptive strategies [4].

Objectives. Each individual is rendered (forward passes only) and scored with the paper's own code: mean per-image CLIPScore as *quality*, and a set-level statistic — patchwise DINOv2 distance, DPP log-det, or Vendi — as *diversity*. NSGA-II maintains the Pareto front of the two, exactly as ECAD maintains quality–versus–speed. Generation 0 is drawn i.i.d. from the prior, so the paper's baseline is embedded in every run as the starting population.

Operators, each borrowed from a finding of theirs. Mutations are spectrally colored by $1/(1{+}f)^{\beta}$, so proposals move low frequencies more than high ones — their Fig. 9

[1] We deliberately avoid the customary term "evolutionary algorithm." Natural evolution optimizes no objective — it is selection without a selector, purposeless by construction — so attaching the word to an explicitly objective-driven procedure seems to us a misnomer. What this algorithm actually performs is closer to what breeders have always done: *artificial selection* toward a stated goal. We therefore speak of *breeding* noise, and of selection-based or population-based search.

Table 1: GenEval, SDXL-Turbo, white-noise initialization, set size 4. Paper rows from Harrington et al. Tab. 1 (their eval harness); our rows use the official repository's own metric code over all 553 prompts in its GenEval list (population 40; up to 60 generations, and prompts still below the reference diversity receive one further 90-generation pass). Diversity metrics: higher is better.

Method	DINO	DreamSim	LPIPS	CLIP
i.i.d.	0.588	0.249	0.642	0.335
Parmar et al. [5]	0.705	0.331	0.682	0.333
Harrington et al. (gradient)	0.784	0.411	0.767	0.349
Bred, gen. 0 (i.i.d.)	0.658	0.298	0.670	0.374
Bred, final	**0.790**	**0.437**	0.745	0.393

Table 2: Design ablations (six GenEval prompts, 45 generations, population 32, matched seeds; defaults marked $\star$; gen-0 i.i.d. reference: DINO 0.639, CLIP 0.338). Variants move *along* the quality–diversity front: the defaults hold the highest quality, while removing the spectral bias (β=0) or fixing a 1% rate is dominated outright.

Group	Variant	DINO	DreamSim	CLIP
mutation rate	fixed 1%	0.705	0.324	0.346
	fixed 15%	0.790	0.464	0.336
	self-tuned $\star$	0.757	0.361	**0.347**
spectral bias	β=0 (white)	0.714	0.316	0.348
	β=1 $\star$	0.757	0.361	0.347
	β=2	**0.830**	**0.593**	0.339
crossover	off	0.802	0.529	0.333
	set-level $\star$	0.757	0.361	0.347
χ_d repair	off	0.750	0.438	0.342
	every op. $\star$	0.757	0.361	0.347
initialization	white $\star$	0.757	0.361	0.347
	pink α=0.2	0.781	0.428	0.341

dynamic, made into an operator (and the same filter behind their pink-noise initialization). Their χ_d-norm regularizer $K(\epsilon)$, which keeps noises in the high-density shell of the prior, becomes a *repair* step: every latent is re-standardized after every operator, enforcing the constraint exactly rather than penalizing its violation. Crossover exchanges whole noise slots between parent sets — recombination in the space of sets, respecting that individual noises are the atoms of diversity.

Self-tuned exploration. In our ECAD ablations, a 1% mutation rate stalls in poor local optima and 15% converges too slowly; the sweet spot was never known. Here each individual carries its own (σ, ρ), and selection tunes them: the population raises σ early (exploration) and anneals it toward 0.05 as it converges (exploitation), with no schedule supplied by us (Fig. 1, right). A controlled comparison (Tab. 2, top) confirms the 1% rate stalls here too, while the aggressive 15% rate buys diversity at a visible quality cost; the self-tuned variant holds the highest quality at comparable diversity, with no rate chosen in advance. The same table shows that removing the low-frequency mutation bias (β=0) loses diversity with no quality gain, while strengthening it (β=2) buys large diversity gains at modest quality cost — both consistent with the source paper's spectral analysis.

3 Results

Table 1 gives the headline. Two sanity checks first: our generation-0 population, scored with the official code, reproduces the paper's i.i.d. row almost exactly, and quality *rises* during search rather than being traded away. Across the reported metrics, breeding lands slightly ahead of the gradient optimizer on DINO, DreamSim, and CLIPScore, and slightly behind on LPIPS — despite never computing $\partial(\text{image})/\partial x_0$. We read this as evidence for a claim implicit in the original paper: the low-frequency structure of the noise landscape is benign enough that *any* sensible search finds it; the gradient is one excellent way, not the only one.

Qualitatively the recovery is the same phenomenon their Fig. 1 shows: four interchangeable grey park benches become a teal bench, a golden-hour park, new viewpoints — every image still plainly a bench (Fig. 2).

Two extensions the population view invites. (i) *Prompt-typed islands.* Sub-populations bred on disjoint prompt distributions (painted landscapes, five-word prompts, long GPT-written prompts), with the best noise sets migrating between islands every few generations. Because a noise set is prompt-agnostic, a migrant that keeps winning after crossing prompt types is one whose diversity *generalizes*: in our runs, migrated elites retain 0.65–0.69 DINO diversity on prompt families they never saw during selection. Per-prompt gradient descent has no natural analogue of this test. (ii) *Searching program space.* Instead of breeding the noise tensor, we can breed a small symbolic *program* (compositions of white noise, pink filtering, and low-frequency gratings) that generates it. Program search reaches the highest raw diversity we observe (DINO 0.886) at a real cost in prompt fidelity — it maps the far end of the quality–diversity front — and its early winners rediscover low-frequency structure on their own, an independent confirmation of the paper's spectral analysis.

4 What each method offers

We want to be precise about the trade, because the comparison flatters neither method alone. The gradient optimizer is far more sample-efficient where it applies: at $0.345\,\text{s}$ per iteration on an A100 (their Sec. 4.2), it reaches its result in seconds per prompt, while breeding spends thou-

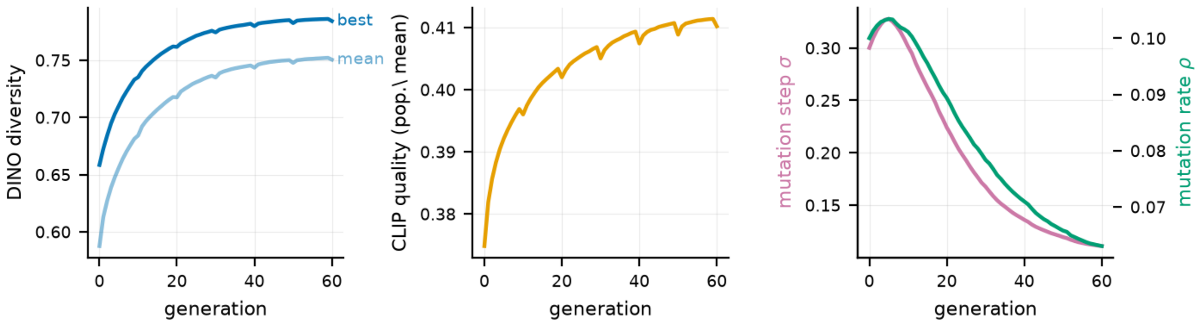

Figure 1: Convergence averaged over GenEval prompts (SDXL-Turbo, population 40). Left: the bred diversity objective. Middle: CLIP quality *rises* during search — the Pareto selection concedes nothing for the diversity gain. Right: the self-adaptive mutation step σ and rate ρ, raised then annealed by selection alone.

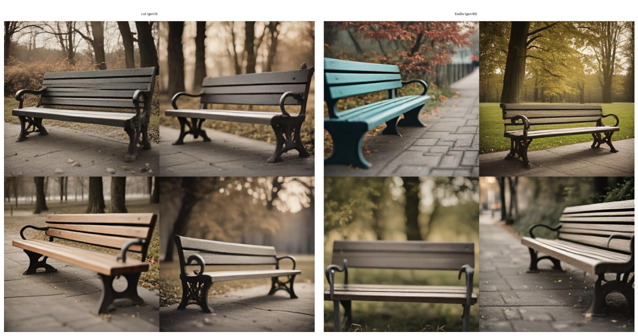

Figure 2: "A photo of a bench," SDXL-Turbo. Left: i.i.d. generation 0. Right: the bred set.

sands of forward renders to arrive at the same neighborhood. Population search earns that cost back in three situations. *Memory:* it never backpropagates through the sampler, so model scale adds no optimizer memory — relevant exactly at the FLUX scale the paper targets. *Objectives:* non-differentiable scores — true DPP samplers, human ratings, detector-based prompt-faithfulness checks — enter the loss with no relaxation, and it is precisely such set-level scores that their user study found most human-aligned. *Output:* a run yields an explicit Pareto *front* of noise sets, so the quality–diversity trade-off becomes a menu rather than a weight to tune. The two approaches also compose naturally: bred noise is a warm start for gradient refinement, and their pink-noise initialization already improves our generation 0 the same way it improves theirs.

5 Closing

Each of the original paper's findings translated directly into a design decision here: the frequency analysis told us what mutations to propose, the norm regularizer told us what constraint to repair to, and the set-level metrics told us what to select for. We offer the population view back as a companion — same landscape, second vehicle — with code, configurations, and scaling scripts in the fork above, in the hope it is useful to the authors' future work on collapse recovery.

References

[1] A. Harrington, A. S. Koepke, S. Karthik, T. Darrell, A. A. Efros. It's Never Too Late: Noise Optimization for Collapse Recovery in Trained Diffusion Models. In *CVPR*, 2026.

[2] A. Aggarwal et al. ECAD: Evolutionary Caching to Accelerate Your Off-the-Shelf Diffusion Model. In *ICLR*, 2026.

[3] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Trans. Evolutionary Computation*, 6(2), 2002.

[4] H.-G. Beyer, H.-P. Schwefel. Evolution strategies: a comprehensive introduction. *Natural Computing*, 1(1), 2002.

[5] G. Parmar et al. Scaling group inference for diverse and high-quality generation. *arXiv:2508.15773*, 2025.